

第十章复习2. 判断

1. (1) $\sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n(n+1)}}$ 条件收敛 $\left\{ \begin{array}{l} \textcircled{1} \frac{1}{\sqrt{n(n+1)}} > 0, \rightarrow 0, \downarrow \\ \textcircled{2} \frac{1}{\sqrt{n(n+1)}} \rightarrow 1, \sum \frac{1}{n} \text{ 发散} \Rightarrow \sum \frac{1}{\sqrt{n(n+1)}} \text{ 发散} \end{array} \right.$

(2) $\sum_{n=2}^{\infty} (-1)^n \frac{1}{n(\ln n)^3}$ 绝对收敛 ~~$\sum_{n=2}^{\infty} (-1)^n \frac{1}{n(\ln n)^3}$~~

$\textcircled{1} \int_2^{\infty} \frac{dx}{x(\ln x)^3} = -\frac{1}{2(\ln x)^2} \Big|_2^{\infty} = \frac{1}{2(\ln 2)^2}$

(3) $\sum_{n=1}^{\infty} (-1)^n \frac{n^2+1}{2n^3+n-1}$ 发散, $\frac{n^2+1}{2n^3+n-1} \rightarrow \frac{1}{2} \neq 0$

(4) $\sum_{n=1}^{\infty} \frac{(-3)^n}{n!}$ 绝对收敛. $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{3}{n+1} = 0 < 1$

2. (1) $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^n$ 绝对收敛 $\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = e^{-1} < 1$

(2) $\sum_{n=1}^{\infty} (-1)^n e^{-\frac{1}{n}}$ 发散. $e^{-\frac{1}{n}} \rightarrow e^0 = 1 \neq 0$

(3) $\sum_{n=2019}^{\infty} (-1)^n \frac{1}{\sqrt{n^2-2019n+1}}$ 条件收敛 $\left\{ \begin{array}{l} \textcircled{1} \frac{1}{\sqrt{n^2-2019n+1}} > 0, \rightarrow 0, \downarrow \\ \textcircled{2} \frac{1}{\sqrt{n^2-2019n+1}} \rightarrow 1, \sum \frac{1}{n} \text{ 发散} \Rightarrow \sum \frac{1}{\sqrt{n^2-2019n+1}} \text{ 发散} \end{array} \right.$

$$3. \cos a_n - \cos b_n = a_n > 0, \quad 0 < a_n < \frac{\pi}{2}, \quad 0 < b_n < \frac{\pi}{2}$$

$$\Rightarrow 0 < a_n < b_n$$

$$\sum b_n \text{ 收敛} \Rightarrow \lim_{n \rightarrow \infty} b_n = 0 \Rightarrow \lim_{n \rightarrow \infty} a_n = 0$$

$$4. \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p (\ln n)^2} \begin{cases} \text{绝对收敛 if } p \geq 1 \\ \text{条件收敛 if } 0 < p < 1 \end{cases}$$

$$\text{if } p > 1, \quad \frac{1}{n^p (\ln n)^2} \leq \frac{1}{n^p}, \quad n \geq 3$$

$$\frac{1}{n^p} \rightarrow 0, \quad \sum \frac{1}{n^p} \text{ 收敛} \Rightarrow \sum \frac{1}{n^p (\ln n)^2} \text{ 收敛}.$$

$$\Rightarrow \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p (\ln n)^2} \text{ 绝对收敛}.$$

$$\text{if } p=1, \quad \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n (\ln n)^2} \text{ 绝对收敛}, \quad \int_2^{\infty} \frac{dx}{x (\ln x)^2} = -\frac{1}{\ln x} \Big|_2^{\infty} = \frac{1}{\ln 2}$$

$$\text{if } 0 < p < 1, \quad \frac{1}{n^p (\ln n)^2} = \frac{n^{1-p}}{(\ln n)^2} \rightarrow \frac{n^{2-p}}{2 \ln n} \rightarrow \infty, \quad \sum \frac{1}{n} \text{ 发散} \Rightarrow \sum \frac{1}{n^p (\ln n)^2} \text{ 发散}$$

$$\text{或 } \frac{1}{n^p (\ln n)^2} > \frac{1}{n^p \cdot n^{2c}} = \frac{1}{n^{p+2c}} \quad (\text{let } 0 < p+2c < 1 \Rightarrow 0 < c < \frac{1-p}{2})$$

$$\Rightarrow \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p (\ln n)^2} \text{ 条件收敛} \quad \text{取 } c = \frac{1-p}{4}$$

$$\frac{1}{n^p (\ln n)^2} > \frac{1}{n^{\frac{3-p}{2}}}, \quad \sum \frac{1}{n^{\frac{3-p}{2}}} \text{ 收敛}$$

5. $c_n = \begin{cases} a^n, & n \text{ is odd} \\ b^n, & n \text{ is even} \end{cases} \quad 0 < b < a < 1$

(a) $\frac{c_n}{\sqrt{(1+a^n)(1+b^n)}} < a^n$, $\sum a^n$ 收敛, $\Rightarrow \sum \frac{c_n}{\sqrt{(1+a^n)(1+b^n)}}$ 收敛.

(b) since $0 < b < a < 1$

$\Rightarrow \left(\frac{a^n}{n^2}\right)^{\frac{1}{n}} < \left(\frac{a^n}{n^2} + \frac{b^n}{n}\right)^{\frac{1}{n}} < \left(\frac{2a^n}{n}\right)^{\frac{1}{n}} \rightarrow a$

$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{a^n}{n^2} + \frac{b^n}{n}\right)^{\frac{1}{n}} = a$

考虑: (1) $\left| \frac{u_{n+1}}{u_n} \right| \rightarrow \left| \frac{(-1)^{n+1}}{(n+2023)^{1/n+1}} \right| < 1 \Rightarrow 0 < x < 2$

when $x=0$, $\sum \frac{(-1)^n}{(n+2023)^{1/n}}$ 收敛

$\frac{1}{n \ln n}$ $\Rightarrow \sum \frac{1}{n \ln n}$ 发散
 $\int_2^{\infty} \frac{dx}{x \ln x} = \ln(\ln x) \Big|_2^{\infty} = \infty$
 ① $\frac{1}{(n+2023)^{1/n}} > 0, \rightarrow 0, \downarrow$

when $x=2$, $\sum \frac{(-1)^n}{(n+2023)^{1/n}}$ 条件收敛. 同上.

$\Rightarrow$ (1) 收敛域 $[0, 2]$

(2) 绝对收敛 $(0, 2)$, 条件收敛, $x=0, x=2$.

$$2. \left| \frac{u_{n+1}}{u_n} \right| < 1 \Rightarrow |x-1| < 1 \Rightarrow 0 < x < 2$$

$$x=2 \quad \sum \frac{1}{n} \text{ 发散}, \quad \frac{\ln n}{n} > \frac{1}{n}, \quad \sum \frac{1}{n} \text{ 发散}.$$

$$x=0 \quad \sum \frac{\ln n}{n} (-1)^n \text{ 条件收敛}, \quad \begin{cases} \frac{\ln n}{n} > 0, \rightarrow 0, \downarrow \\ \frac{\ln n}{n} > \frac{1}{n}, \sum \frac{1}{n} \text{ 发散} \end{cases}$$

$$\Rightarrow (0, 2) \quad (2) \text{ 绝对收敛} \quad (3) \text{ 条件收敛} \cdot x=0.$$

$$3. \left| \frac{u_{n+1}}{u_n} \right| < 1 \Rightarrow |x| < 1 \Rightarrow -1 < x < 1.$$

$$x=1, \quad \sum \frac{n!}{n(n+1)} \text{ 发散}, \quad \frac{n!}{n(n+1)} \rightarrow 1, \quad \sum \frac{1}{n} \text{ 发散}.$$

$$x=-1, \quad \sum (-1)^n \frac{n!}{n(n+1)} \text{ 条件收敛} \quad \begin{cases} \frac{n!}{n(n+1)} > 0, \rightarrow 0, \downarrow \\ \frac{n!}{n(n+1)} \rightarrow 1, \sum \frac{1}{n} \text{ 发散} \end{cases}$$

$$\Rightarrow [-1, 1)$$

$$(2) \sum_{n=1}^{\infty} \frac{n+2}{n(n+1)} x^n = \sum_{n=1}^{\infty} \frac{1}{n+1} x^n + 2 \sum_{n=1}^{\infty} \frac{1}{n(n+1)} x^n$$

$$= \sum_{n=1}^{\infty} \frac{1}{n+1} x^n + 2 \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) x^n$$

$$= 2 \sum_{n=1}^{\infty} \frac{1}{n} x^n - \sum_{n=1}^{\infty} \frac{1}{n+1} x^n$$

$$\left(\sum_{n=1}^{\infty} \frac{x^n}{n} \right)' = \sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x) \quad x \neq 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n+1} x^n = \sum_{n=1}^{\infty} \frac{1}{n+1} x^{n+1} \cdot \frac{1}{x} = \frac{1}{x} \sum_{n=1}^{\infty} \frac{1}{n+1} x^{n+1}$$

$$\left(\sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} \right)' = \sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} = -\ln(1-x) - x \quad x \neq 0$$

$$4. \left| \frac{u_{n+1}}{u_n} \right| = 2|x| < 1 \Rightarrow -\frac{1}{2} < x < \frac{1}{2}$$

$$x=\frac{1}{2}, \quad \sum \frac{1}{\ln(n+2)} \text{ 发散}, \quad \frac{1}{\ln(n+2)} > \frac{1}{n+2}, \quad \sum \frac{1}{n+2} \text{ 发散}.$$

$$x=-\frac{1}{2}, \quad \sum (-1)^n \frac{1}{\ln(n+2)} \text{ 条件收敛} \quad \begin{cases} \frac{1}{\ln(n+2)} > 0, \rightarrow 0, \downarrow \\ \frac{1}{\ln(n+2)} > \frac{1}{n+2}, \sum \frac{1}{n+2} \text{ 发散} \end{cases}$$

$$\Rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right)$$

$$-\frac{1}{2} < x < \frac{1}{2}$$

$$x=\frac{1}{2}, \quad \sum \frac{(-1)^n}{\sqrt{n+1}} \text{ 条件收敛}$$

$$\begin{cases} \frac{1}{\sqrt{n+1}} > 0, \rightarrow 0, \downarrow \\ \frac{1}{\sqrt{n+1}} > \frac{1}{n}, \sum \frac{1}{n} \text{ 发散} \end{cases}$$

$$5. \left| \frac{u_{n+1}}{u_n} \right| < 1 \Rightarrow |x-1|^2 < 1 \Rightarrow 0 < x < 2 \Rightarrow (2) \text{ 绝对收敛}$$

$$x=2 \quad \sum \frac{(-1)^n}{\sqrt{n+1/2} \ln n} \text{ 条件收敛} \quad \begin{cases} \frac{1}{\sqrt{n+1/2} \ln n} > 0, \rightarrow 0, \downarrow \\ \frac{1}{\sqrt{n+1/2} \ln n} > \frac{1}{\sqrt{n+1/2}}, \sum \frac{1}{\sqrt{n+1/2}} \text{ 发散} \end{cases}$$

$$x=0, \quad \sum \frac{(-1)^n}{\sqrt{n+1/2} \ln n} \text{ 条件同上} \rightarrow \frac{1}{\sqrt{n+1/2} \ln n} > \frac{1}{\sqrt{n+1/2} \cdot \sqrt{n}} > \frac{1}{n+1/2}, \quad \sum \frac{1}{n+1/2} \text{ 发散}$$